Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

In review ratings column

Key features include demographics, purchase details, and shopping behavior metrics.

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Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Checked structure and summary statistics

03

Missing Data Handling

Imputed missing review ratings by category

04

Feature Engineering

Created age groups and purchase frequency metrics

05

Database Integration

Loaded cleaned data into PostgreSQL

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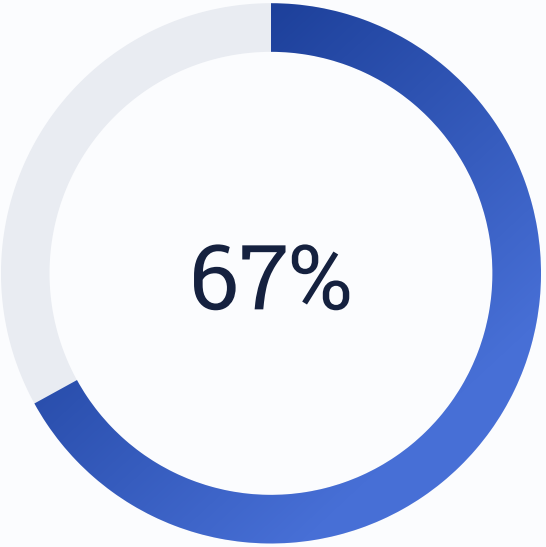
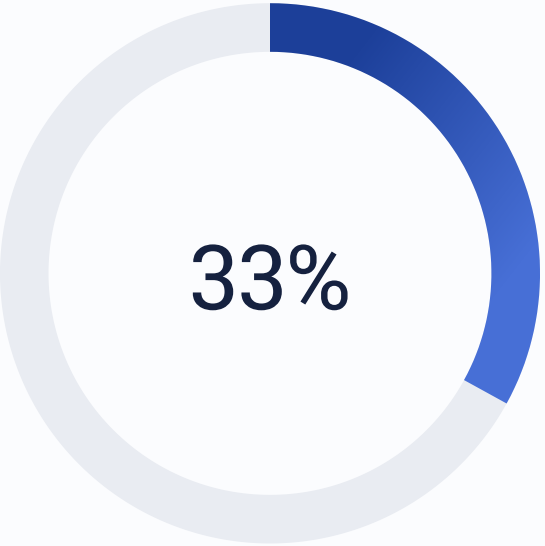
Revenue by Gender



Male Customers

$157,890 total revenue



Female Customers

$75,191 total revenue

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Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

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Shipping & Subscription Insights

Shipping Comparison

- Standard: $58.46 avg purchase
- Express: $60.48 avg purchase

Subscription Status

- Non-subscribers: 73% (2,847 customers)
- Subscribers: 27% (1,053 customers)

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Customer Segmentation

Loyal Customers

3,116 customers with consistent purchase history

Returning Customers

701 customers making repeat purchases

New Customers

83 customers with single transactions

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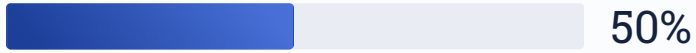
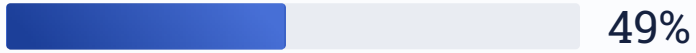
Discount Strategy Analysis



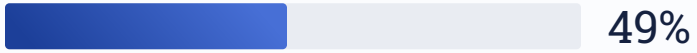
Hats

Most discounted product



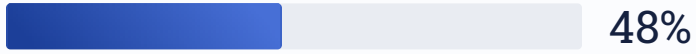
Sneakers

High discount rate



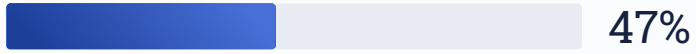
Coats

Seasonal promotions



Sweaters

Fashion category



Pants

Core clothing item

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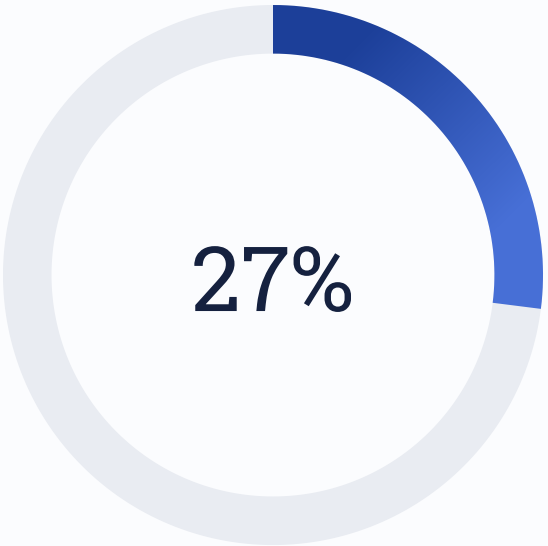
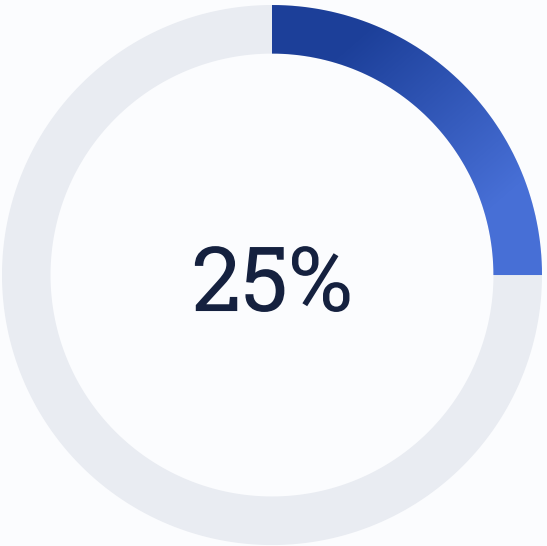
Revenue by Age Group



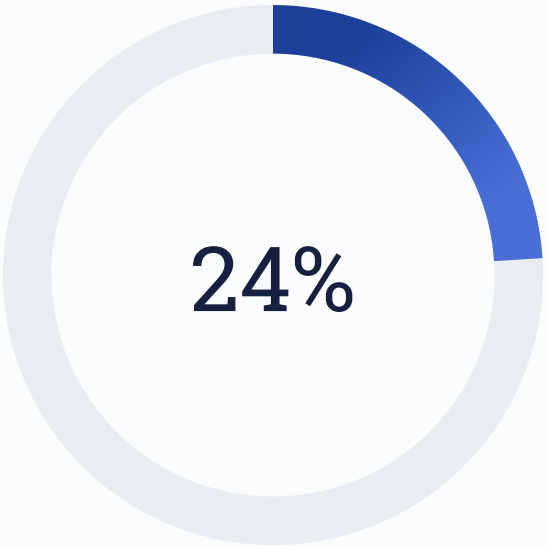
Young Adult

$62,143 revenue



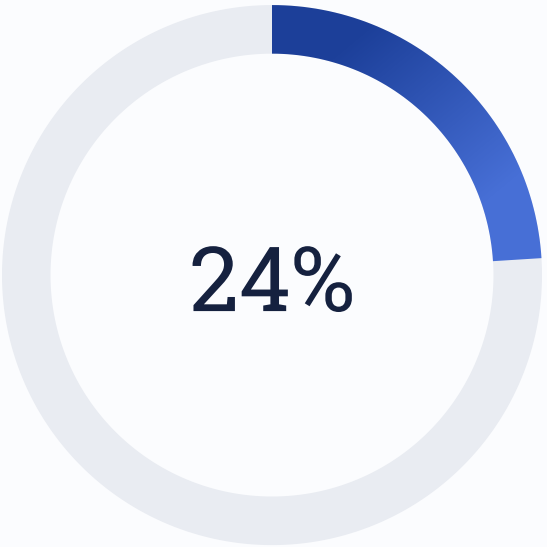
Middle-aged

$59,197 revenue



Adult

$55,978 revenue



Senior

$55,763 revenue

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber base

Loyalty Programs

Reward repeat buyers to build loyal customer segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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